21/08/26

Module - 03

Probability Distribution :-

$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ $f(x) = x^2$ $\begin{pmatrix} 1 \\ 4 \\ 9 \end{pmatrix}$

domain co-domain

$\begin{pmatrix} HH \\ HT \\ TH \\ TT \end{pmatrix}$ X No. of head $\begin{pmatrix} 2 \\ 1 \\ 1 \\ 0 \end{pmatrix}$

Sample Space Real no.

x	0	1	2
$P(x)$	$1/4$	$1/2$	$1/4$

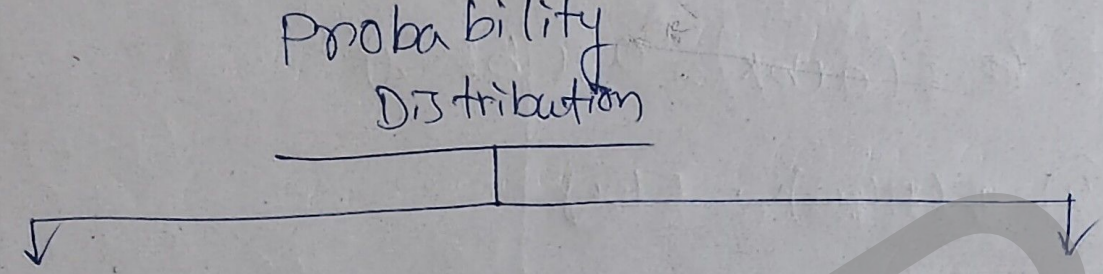
Random variable (RV) :-

Def :- A RV is a real valued function define over a sample space Ω. It is denoted by x or $x(\omega)$

$\{x(\omega) : \omega \in \Omega, x(\omega) \in \mathbb{R}\}$

(11) Distribution :-

A RV together with its corresponding probability is known as distribution of RV x.



Discrete

1. Binomial.
2. Poisson
3. Geometric

continuous

1. Uniform
2. Normal
3. Exponential.

(12) Expectation :-

Mean of a distribution

$$E(x) = \sum_{x=-\infty}^{\infty} x \cdot p(x) \quad [\text{Discrete RV}]$$

$$E(x) = \int_{-\infty}^{\infty} x \cdot p(x)\, dx \quad [\text{cont. RV}]$$

$$\therefore E(x) = 0 \times \frac{1}{4} + 1 \times \frac{2}{4} + 2 \times \frac{1}{4}$$

$$= 0 + \frac{2}{4} + \frac{2}{4} = 1$$

⑪ Properties:-

① $E(a) = a$

② $E(bx) = bE(x)$

③ $E(a+bx) = a + bE(x)$

④ $E(E(x)) = E(x)$

* $E(x) = 2$
  $E(7+3x) = 13$

⑫ Variance:-

$$\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

$$Var(x) = E(x - E(x))^2$$

$$= E\{x^2 - 2xE(x) + (E(x))^2\}$$

$$= E(x^2) - 2(E(x))^2 + (E(x))^2$$

$$Var(x) = E(x^2) - (E(x))^2$$

$E(X^2) = \sum_{\text{all } x} x^2 \cdot P(x)$

x	0	1	2
$P(x)$	$1/4$	$1/2$	$1/4$

$= 0^2 \times \frac{1}{4} + 1 \times \frac{1}{2} \times 4 \times \frac{1}{4}$

$= 0 + 1/2 + 1$

$= 3/2$

$\therefore E(x) = 1$ [Proved before]

$\therefore Var(x) = 3/2 - 1$

$= 1/2$

Variance properties:—

1. $Var(a) = 0$
2. $Var(bx) = b^2 Var(x)$
3. $Var(a+bx) = b^2 Var(x)$
4. $Var(x+y) = Var(x) + Var(y) - 2Cov(x,y)$

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① Probability Function :-

Discrete — SAQ 1

Probability Mass Function (PMF)

Def:- The PMF of a RV is denoted by $f(x)$, where, $f(x) = P(X = x)$

properties:-

① $f(x) \geq 0 \quad \forall x$

② $\sum_{x} f(x) = 1$

continuous — SAQ 2

probability Density Function (PDF)

Def:- The PDF of a RV is denoted by $f(x)$, where,

$$\int_a^b f(x)\,dx = P(a \leq x \leq b)$$

properties:-

① $f(x) \geq 0 \quad \forall x$

② $\int_{-\infty}^{\infty} f(x)\,dx = 1$

S.A.Q-3

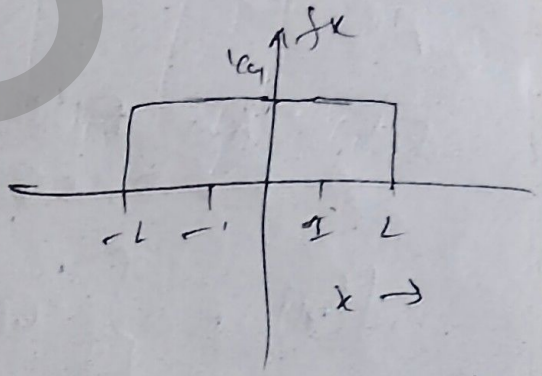
If a RV x has the PDF $f(x) = \begin{cases} \frac{1}{4}, & -2 < x < 2 \\ 0, & \text{otherwise} \end{cases}$

Identify the name of $P\{(2x+3) > 5\}$?

Ans :=>

P.T.O

2. $P\{(2x+3) > 5\}$



$= P(2x+3 > 5)$

$= P(2x > 5-3)$

$= P(2x > 2)$

$= P(x > 1)$

$\therefore \int_1^2 f(x)\,dx$

$= \int_1^2 \frac{1}{4}\,dx$

$= \frac{1}{4}[x]_1^2$

$= \frac{1}{4}[2-1]$

$= \frac{1}{4}$

L.AQ-1>

Q.) The PDF of X is given by

$$f(x) = \begin{cases} \frac{3}{4}x(2-x), & 0<x<2 \\ 0, & \text{otherwise} \end{cases}$$

Find (i) $E(x)$

(ii) $P(x<1)$

Ans:-

$$E(x) = \int_{-\infty}^{\infty} x \cdot f(x)\, dx.$$

$$= \int_0^2 x \cdot \left\{\frac{3}{4}x(2-x)\right\} dx$$

$$= \frac{3}{4}\int_0^2 x^2(2-x)\, dx$$

$$= \frac{3}{4}\left\{\int_0^2 2x^2\, dx - \int_0^2 x^3\, dx\right\}$$

$$= \frac{3}{4}\left[\frac{x^3}{3}\right]_0^2 - \frac{3}{4}\left[\frac{x^4}{4}\right]_0^2$$

$$= \frac{1}{4}\left[x^3\right]_0^2 - \frac{3}{16}x$$

$$= \left[\frac{3}{4} \times 2 \times \frac{1}{3} x^3\right]_0^2 - \left[\frac{3}{16} x^4\right]_0^2$$

$$= \frac{1}{2} \times [8-0] - \frac{3}{16} \times [16-0]$$

$$= \frac{1}{2} \times 8^4 - \frac{3}{16} \times 16$$

$$= 4 - 3 = 1 \quad \text{(Ans)}$$

(ii) $P(x < 1)$

$$= \int_0^1 f(x)\, dx$$

$$= \int_0^1 \frac{3}{4} x(2-x)$$

$$= \frac{3}{4} \int_0^1 (2x - x^2)\, dx$$

$$= \frac{3}{4} \left\{ \int_0^1 2x\, dx - \int_0^1 x^2\, dx \right\}$$

$$= \left[\frac{3}{4} \times 2 \times \frac{1}{2} \times x^2\right]_0^1 - \frac{3}{4} \left[\frac{1}{3} \times x^3\right]_0^1$$

$$= \frac{3}{4} \times 2 \times \frac{1}{2} \times 1 - \frac{3}{4} \times \frac{1}{3}$$

$$= \frac{3}{4} - \frac{1}{4}$$

$$= \frac{2}{4} = \frac{1}{2}$$

S.A.Q 8

A RV x has the PMF:

x	0	1	2	3
P(X=x)	K	2K	3K	4K
	1/10	4/10	3/10	4/10

Calculate – (i) K
(ii) E(x)
(iii) var(x)

Ans:

(i) By function

$$\sum_{x=0}^{3} f(x) = 1$$

$\Rightarrow K+2K+3K+4K = 1$

$\Rightarrow K = \frac{1}{10}$

(ii)

$E(x) = \sum x P(x)$

$= 0 \times \frac{1}{10} + 1 \times \frac{2}{10} + 2 \times \frac{3}{10} + 3 \times \frac{4}{10}$

$= \frac{2}{10} + \frac{6}{10} + \frac{12}{10} = \frac{20}{10}$

$= 2$

$$Var(x) = E(x^2) - (E(x))^2$$

$$\therefore E(x^2) = \sum x^2 \cdot p(x)$$

$$= 0 + \frac{2}{10} + \frac{12}{10} + \frac{36}{10}$$

$$= \frac{50}{10} = 5$$

$$\therefore Var(x) = 5 - 4$$

$$= 1$$

(ii) Covariance:—

$$\frac{1}{n} \sum (x_i - \bar{x})(y_i - \bar{y})$$

$$cov(x,y) = E\{(x - E(x))(y - E(y))\}$$

$$= E\{xy - xE(y) - E(x)x + E(x)E(y)\}$$

$$= E(xy) - E(x)E(y) - \cancel{E(x)E(y)} + \cancel{E(x)E(y)}$$

$$\boxed{cov(x,y) = E(xy) - E(x)E(y)}$$

(11) Correlation coefficient :-

$$\rho_{xy} = \frac{cov(x,y)}{S_x S_y} = \frac{E(XY) - E(X)E(Y)}{\sqrt{E(X^2) - (E(X))^2} \times \sqrt{E(Y^2) - (E(Y))^2}}$$

(*) If x and y are independent !

$$E(XY) = E(X)E(Y)$$

SAQ -10)

A fair coin is tossed 3 times.

X = 0 if 1st toss gives tail

X = 1 if 1st toss gives head

Y = No. of head obtained in 3 tosses

Find the joint - distribution table of X and Y.

=> Ω = {HHH, HHT, HTH, THH, TTH, HTT, THT, TTT}

x \ y	0	1	2	3
(T) 000	1/8	2/8	1/8	0
(H) 00.1	0	2/8	2/8	1/8

H.W

S.A-Q (17)

$$f(x) = \begin{cases} \frac{1}{6}, & 2 < x < 8, \\ 0, & \text{otherwise} \end{cases}$$

(i)

$$\therefore P(3 < x < 6) = \int_3^6 f(x)\,dx$$

$$= \int_3^6 \frac{1}{6}\,dx$$

$$= \frac{1}{6}[x]_3^6$$

$$= \frac{1}{6} \times 3 = \frac{1}{2}$$

(ii) $E(x) = \int_2^8 x f(x)\,dx$

$$= \int_2^8 \frac{x}{6}\,dx$$

$$= \frac{1}{6}\left[\frac{x^2}{2}\right]_2^8$$

$$= \frac{1}{12}(64-4)$$

$$= \frac{1}{12} \times 60 = 5$$

A-9

(6)

$$f(x) = \begin{cases} cx, & 0 \le x \le 1 \\ c, & 1 \le x < 2 \\ -cx + 3c, & 2 \le x \le 3 \\ 0, & \text{otherwise} \end{cases}$$

∴ Since total prob = 1

$$\therefore \int_0^3 f(x)\,dx = 1$$

$$\Rightarrow \int_0^1 cx\,dx + \int_1^2 c\,dx + \int_2^3 (-cx+3c)\,dx = 1$$

$$\Rightarrow c\left[\frac{x^2}{2}\right]_0^1 + c[x]_1^2 - c\left[\frac{x^2}{2}\right]_2^3 + 3c[x]_2^3 = 1$$

$$\Rightarrow \frac{c}{2} \times 1 + c - \frac{c}{2} \times 5 + 3c \times 1 = 1$$

$\Rightarrow \frac{c}{2} + c - \frac{c}{2} + 3c = 1$

$\Rightarrow 2c = 1$

$\Rightarrow c = 1/2$

$\therefore E(x) = \int_0^3 x f(x)\,dx$

$= \int_0^1 cx^2\,dx + \int_1^2 cx\,dx + \int_2^3 -cx^2\,dx + \int_2^3 3cx\,dx$

$= c\left[\frac{x^3}{3}\right]_0^1 +$

$\Rightarrow c\left[\frac{x^2}{2}\right]_0^1 + c[x]_1^2 - c\left[\frac{x^2}{2}\right]_2^3 + 3c[x]_2^3$

$= 1$

$\Rightarrow \frac{c}{2} \times \frac{1}{2} + c \times 1 - c \times \frac{5}{2} + 3c = 1$

$\Rightarrow \frac{c}{2} + c - \frac{5c}{2} + 3c = 1$

$\Rightarrow \frac{c + 2c - 5c + 6c}{2} = 1$

$\Rightarrow 2c = 1 \Rightarrow c = 1/2$

$\therefore E(x) = \int_0^3 x f(x)\, dx$

$= \int_0^1 cx^2\, dx + \int_1^2 cx\, dx - \int_2^3 cx^2\, dx + \int_2^3 3c\, dx$

$= c\left[\frac{x^3}{3}\right]_0^1 + c\left[\frac{x^2}{2}\right]_1^2 - c\left[\frac{x^3}{3}\right]_2^3 + 3c\left[\frac{x^2}{2}\right]_2^3$

$= c \times \frac{1}{3} + c \times \frac{3}{2} - c\,\frac{19}{3} + 3c \times \frac{5}{2}$

$= \frac{c}{3} + \frac{3c}{2} - \frac{19c}{3} + \frac{15c}{2}$

$= \frac{18c}{2} - \frac{18c}{3}$

$= \frac{54c - 36c}{6}$

$= \frac{18c}{6}$

$= \frac{18}{6} \times \frac{1}{2}$

$\therefore E(x) = 3/2$

$\therefore \text{Var}(x) = E(x^2) - [E(x)]^2$

$$\therefore E(x^2) = \int_0^3 x^2 f(x)\,dx$$

$$= \int_0^1 x^2(cx)\,dx + \int_1^2 x^2 c\,dx + \int_2^3 x^2(-cx+3c)\,dx$$

$$= \int_0^1 cx^3\,dx + \int_1^2 cx^2\,dx - \int_2^3 cx^3\,dx + 3\int_2^3 cx^2\,dx$$

$$= c\left[\frac{x^4}{4}\right]_0^1 + c\left[\frac{x^3}{3}\right]_1^2 - c\left[\frac{x^4}{4}\right]_2^3 + 3c\left[\frac{x^3}{3}\right]_2^3$$

$$= c \times \frac{1}{4} + c \times \frac{7}{3} - c \times \frac{65}{4} + 3c \times \frac{19}{3}$$

$$= \frac{7c}{3} + \frac{57c}{3} - \frac{64c}{4}$$

$$= \frac{64c}{3} - 16c$$

$$= \frac{16c}{3}$$

$$= \frac{16}{3} \times \frac{1}{2} = \frac{8}{3}$$

$$\therefore \text{Var}(X) = \frac{8}{3} - \frac{9}{4}$$

$$= \frac{32 - 27}{12} = \frac{5}{12}$$

Good for 24

28/08/26
8.A-9-1

Cumulative Distribution Function (CDF):-

A CDF is denoted by $F(x)$,

where, $F(x) = P(x \leq x)$

Properties

(1) $F(+\infty) = 1$

(2) $F(-\infty) = 0$

L.A-9

(15) If x represents the outcome in rolling a dice calculate $F(x)$

Ans:-

x	1	2	3	4	5	6
$P(x)$	1/6	1/6	1/6	1/6	1/6	1/6
$F(x)$	1/6	2/6	3/6	4/6	5/6	6/6 = 1

Discrete Distribution:-

Bernoulli Trial:-

1> The Trial must have two outcomes success and failure.

2> All the trials must be independent.

3) probability of success for each trial should remain same.

If n = no. of trial = 1 ⟶ Bernoulli distribution

$n > 1$ ⟶ Binomial ,,

* **Binomial Distribution:-**

$$\text{PMF}: f(x) = \begin{cases} {}^{n}C_{x}\, p^{x}(1-p)^{n-x}, & x = 0, 1, 2 \ldots n \\ 0, & \text{otherwise.} \end{cases}$$

n = no. of trial

x = no. of success.

p = probability of success

$q = (1-p)$ = prob of failure

Notation: $\boxed{X \sim \text{Bin}(n, p)}$

$n = 50$, $p(\text{head}) = \frac{1}{2}$

$$p(x=30) = {}^{50}C_{30}\left(\frac{1}{2}\right)\left(1-\frac{1}{2}\right)^{50-30}$$

S.A.Q-4

Show that mean of Bin(n,p) is np?

$$\Rightarrow E(x) = \sum_{x=-\infty}^{\infty} x \cdot f(x)$$

$$= \sum_{x=0}^{n} x \cdot {}^{n}C_x p^x (1-p)^{n-x}$$

$$= \sum_{x=0}^{n} x \cdot \frac{n!}{(n-x)!\,x!} p^x (1-p)^{n-x}$$

$$= \sum_{x=1}^{n} \frac{n!}{(n-x)!(x-1)!} p^x (1-p)^{n-x}$$

$$= np \sum_{x=1}^{n} \frac{(n-1)!}{(n-x)!(x-1)!} p^{x-1} (1-p)^{n-x}$$

$$= np \sum_{y=x-1=0}^{n-1} \frac{(n-1)!}{(n-y-1)!\,y!} p^y (1-p)^{n-y-1}$$

$$= np \sum_{y=0}^{n-1} {}^{n-1}C_y p^y (1-p)^{(n-1)-y}$$

$$= np \left[\because f(y) = \sum_{y=0}^{n-1} {}^{n-1}C_y \, p^y (1-p)^{(n-1)-y} \right]$$

$\sim$ Binomial

$\sum f(y) = 1$]

01/09/26

ⓘ Binomial Variance! —

L.A-Q -02

$$\text{Var}(x) = E(x^2) - (E(x))^2$$

$$= E(x(x-1)) + E(x) - (E(x))^2 \quad [\text{Discrete}]$$

$$E(x(x-1)) = \sum_{x=0}^{n} x(x-1) \cdot f(x)$$

$$= \sum_{x=0}^{n} x(x-1) \cdot {}^{n}C_x \, p^x (1-p)^{n-x}$$

$$= \sum_{x=0}^{n} x(x-1) \frac{n!}{(n-x)!\, x!} p^x (1-p)^{n-x}$$

$$= \sum_{x=2}^{n} \frac{n!}{(n-x)!\,(x-2)!} p^x (1-p)^{n-x}$$

$$= n(n-1)p^2 \sum_{x=2}^{n} \frac{(n-2)!}{(n-x)!\,(x-2)!} p^{(x-2)} (1-p)^{n-x}$$

$$= n(n-1)p^2 \sum_{y=x-2=0}^{n-2} \frac{(n-2)!\; p^y (1-p)^{(n-y-2)}}{(n-y-2)!\; y!}$$

$$= n(n-1)p^2 \sum_{y=0}^{n-2} {}^{(n-2)}C_y\, p^y (1-p)^{(n-y-2)}$$

$$= n(n-1)p^2 \times 1 \quad \left[\because \sum_{y=0}^{n-2} {}^{n-2}C_y\, p^y (1-p)^{n-2-y} = f(y) \right.$$

Binomial sum

$$\text{Var}(x) = n(n-1)p^2 + np - n^2p^2$$

$$= n^2p^2 - np^2 + np - n^2p^2$$

$$= np(1-p)$$

$$= npq.$$

L.A-9
17

Given,

Mean $(E(x)) = 4$, $E(x^2) = 19$

$x \sim Bin(n, p)$

$E(x) = 4$

$np = 4$ ——— (1)

$\therefore Var(x) = E(x^2) - (E(x))^2$

$= 19 - 16 = 3$

$\therefore npq = 3$

$\Rightarrow np(1-p) = 3$

$\Rightarrow 4(1-p) = 3$

$\Rightarrow 1-p = \frac{3}{4}$

$\Rightarrow p = 1 - \frac{3}{4}$

$\Rightarrow p = \frac{1}{4}$

From, ——— (1)

$n \times \frac{1}{4} = 4 \Rightarrow n = 16$

NOW,

$X \sim$ Binomial

$f(x) = {}^{n}C_x \, p^x (1-p)^{n-x}$

$P(x=3) = {}^{16}C_3 \left(\frac{1}{4}\right)^3 \left(1-\frac{1}{4}\right)^{16-3}$

$= {}^{16}C_3 \times \left(\frac{1}{4}\right)^3 \left(\frac{3}{4}\right)^{13}$

$= 0.207$

$P(x = \text{at most } 2)$

$= P(x \leq 2)$

$= P(x=0) + P(x=1) + P(x=2)$

$\therefore P(x=0) = {}^{16}C_0 \left(\frac{1}{4}\right)^0 \times \left(1-\frac{1}{4}\right)^{16-0}$

$= {}^{16}C_0 \times \left(\frac{3}{4}\right)^{16}$

$= 0.01$

$$P(x=1) = {}^{16}C_1 \times \left(\frac{1}{4}\right)^1 \times \left(1-\frac{1}{4}\right)^{16-1}$$

$$= 0.053$$

$$P(x=2) = {}^{16}C_2 \left(\frac{1}{4}\right)^2 \times \left(\frac{3}{4}\right)^{14}$$

$$= 0.133$$

$$\therefore P(x \leq 2) = 0.01 + 0.053 + 0.133$$

$$= 0.196.$$

H.W

L.A-9

16

S.A.Q 11

Given, probability $(p) = 0.1$

X: No. of defective items.

$X \sim Bin(n=8, p=0.1)$

$$f(x) = {}^8C_x (0.1)^x (1-0.1)^{8-x}$$

$$= {}^8C_x (0.1)^x (0.9)^{8-x}$$

$\therefore P(x \leq 2)$

$= P(x=0) + P(x=1) + P(x=2)$

$= {}^8C_0 (0.1)^0 (0.9)^8 + {}^8C_1 (0.1)^1 (0.9)^7 + {}^8C_2 (0.1)^2 (0.9)^6$

$= 0.43 + 0.38 + 0.15$

$= 0.96.$

* Mode:—

$$K = (n+1)p$$

If the value of K is an integer

$$\text{mode} = (K-1), K$$

If the value of K is a fraction.

$$\text{mode} = [K]$$

Example:

$$X \sim \text{Bin}(n=16, p=\tfrac{1}{4})$$

$$K = (16+1) \times \tfrac{1}{4}$$

$$= \frac{17}{4}$$

$$= 4.25$$

$$\text{mode}[4.25] = 4$$

S.A-9 / 7

Given,

$$\text{mean} = 5/3$$

$$P(x=2) = P(x=1)$$

$$\therefore E(x) = 5/3$$

$$\Rightarrow np = 5/3 \quad \text{——} (1)$$

$\therefore P(x=2) = P(x=1)$

$\Rightarrow {}^{n}C_2 p^2 (1-p)^{n-2} = {}^{n}C_1 p (1-p)^{n-1}$

$\Rightarrow \frac{n!}{2!(n-2)!} p = n(1-p)$

$\Rightarrow \frac{n(n-1)}{2} p = n(1-p)$

$\Rightarrow np - p = 2 - 2p$

$\Rightarrow np = 2 - p$

$\Rightarrow 5/3 = 2 - p$

$\Rightarrow p = 2 - 5/3$

$\Rightarrow p = \frac{1}{3}$

$\therefore np = 5/3$

$\Rightarrow n = 5/3 \times 3$

$\Rightarrow n = 5$

$\therefore K = (5+1) \cdot \frac{1}{3}$

$= 6 \times \frac{1}{3} = 2$

mode = (1, 2)

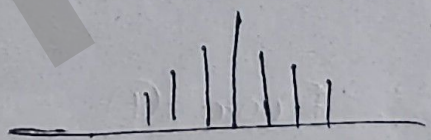
Variance $= npq(1-p)$

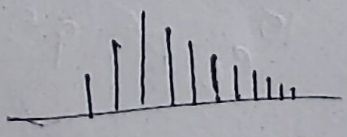
$= 5 \times \frac{1}{3} \left(1 - \frac{1}{3}\right)$

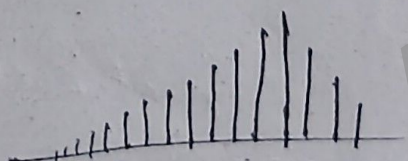
$= 5 \times \frac{1}{3} \times \frac{2}{3}$

$= \frac{10}{9}$

⓪ Skewness

$p = 1/2$ (Symmetrical) 

$p < 1/2$ (positively skewed) 

$p > 1/2$ (Negatively skewed) 

⓪ Kurtosis:-

$pq = \frac{1}{6}$ (Meso)

$pq < \frac{1}{6}$ (Lepto)

$pq > 1/6$ (platy)

Ans

L.A.Q

16

Given,

$$P(x=5) = 2P(x=4) \quad \text{---} (1)$$

$$x \sim Bin(n=10, P)$$

$\therefore$ we know,

$$q = 1-p$$

$$\therefore P(x=5) = {}^{10}C_5 p^5 q^5$$

$$P(x=4) = {}^{10}C_4 p^4 q^6$$

$\therefore$ From (1)

$${}^{10}C_5 p^5 q^5 = 2 \times {}^{10}C_4 p^4 q^6$$

$$\Rightarrow 252 \times p = 2 \times 210 \times q$$

$$\Rightarrow \frac{p}{q} = \frac{420}{252}$$

$$\Rightarrow 252p = 420q$$

$$\Rightarrow 252p = 420 - 420p$$

$$\Rightarrow 672p = 420 \Rightarrow p = \frac{420}{672}$$

$$= 5/8$$

$\therefore q = 1 - \frac{5}{8}$

$= 3/8$

$\therefore P(x=1)$

$= {}^{10}C_1 \left(\frac{5}{8}\right) \times \left(\frac{3}{8}\right)^9$

$= 0.00091$

$\therefore P(x>0 / x \leq 2)$

~~$= P(x=1) + P(x=2)$~~

$= \dfrac{P(x=1) + P(x=2)}{P(x=0) + P(x=1) + P(x=2)}$

$\therefore P(x=0) = \left(\frac{3}{8}\right)^{10} = 0.000054$

$P(x=1) = 0.00091$

$P(x=2) = {}^{10}C_2 \left(\frac{5}{8}\right)^2 \left(\frac{3}{8}\right)^8$

$= 0.0068$

$\therefore P(x>0 / x \leq 2)$

$= \dfrac{0.00091 + 0.0068}{0.000054 + 0.00091 + 0.0068}$

$= \dfrac{0.00771}{0.00776} = 0.994$ (Ans)

⑪ Poisson Distribution:-

PMF:-

$$f(x) = \begin{cases} e^{-\lambda}\frac{\lambda^x}{x!}, & x = 0, 1, 2, \ldots \infty \\ 0, & \text{otherwise} \end{cases}$$

λ = Rate of occurance of event

Notation:-

$$x \sim \text{poisson}(\lambda)$$

S.A-Q-6)

Mean: $E(x) = \sum_{x=0}^{\infty} x \cdot f(x)$

$$= \sum_{x=0}^{\infty} x \cdot e^{-\lambda}\frac{\lambda^x}{x(x-1)!}$$

$$= \sum_{x=1}^{\infty} e^{-\lambda}\frac{\lambda^x}{(x-1)!}$$

$$= \lambda e^{-\lambda} \sum \frac{\lambda^{x-1}}{(x-1)!}$$

$$= \lambda e^{-\lambda}\left\{1 + \lambda + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} + \ldots \infty\right\}$$

$$= \lambda e^{-\lambda} \cdot e^{\lambda} = \lambda$$

L.A.Q-4 Also calculate mean

Q. variance:-

$$E(x(x-1)) + E(x) - (E(x))^2$$

$$\therefore E(x(x-1)) = \sum_{x=0}^{\infty} x(x-1) f(x)$$

$$= \sum_{x=0}^{\infty} x(x-1) \cdot e^{-\lambda} \frac{\lambda^x}{x!}$$

$$= \sum_{x=2}^{\infty} e^{-\lambda} \frac{\lambda^x}{(x-2)!}$$

$$= \lambda^2 e^{-\lambda} \sum_{x=2}^{\infty} \frac{\lambda^{x-2}}{(x-2)!}$$

$$= \lambda^2 e^{-\lambda} \left\{ 1 + \lambda + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} + \cdots \infty \right\}$$

$$= \lambda^2 e^{-\lambda} \cdot e^{\lambda}$$

$$= \lambda^2$$

$$\therefore \text{Variance} = \lambda^2 + \lambda - \lambda^2$$

$$= \lambda$$

S.A-9
14

Given,

$E(x) = 1.5 = \lambda$

~~$x \sim$ possion (λ)~~

x: No. of typing errors per page

$x \sim$ poisson ($\lambda = 1.5$)

$\therefore f(x) = e^{-1.5} \frac{(1.5)^x}{x!}$

(i) $P(x=0) = e^{-1.5} \frac{(1.5)^0}{0!}$

$= 0.223$

(ii) $P(x \geq 2) = 1 - P(x < 2)$

$\Rightarrow P(x \geq 2) = 1 - \{P(x=0) + P(x=1)\}$ ——— (1)

$\therefore P(x=1) = e^{-1.5} \frac{(1.5)^1}{1!}$

$= e^{-1.5} \times 1.5$

$= 0.334$

$\therefore P(x \geq 2)$

From — (ii)

$P(x \geq 1) = 1 - 0.223 = 0.334$

$= 0.443$

S.A.Q 18

Given,

x: customers per hour

$x \sim$ poisson ($\lambda = 4$)

$\therefore E(x) = 4$

$\therefore f(x) = e^{4} \frac{(4)^x}{x!}$

$\therefore P(x=5) = e^{-4} \frac{(4)^5}{5!}$

$= 0.156$

08/09/26

* Continuous Distribution:-

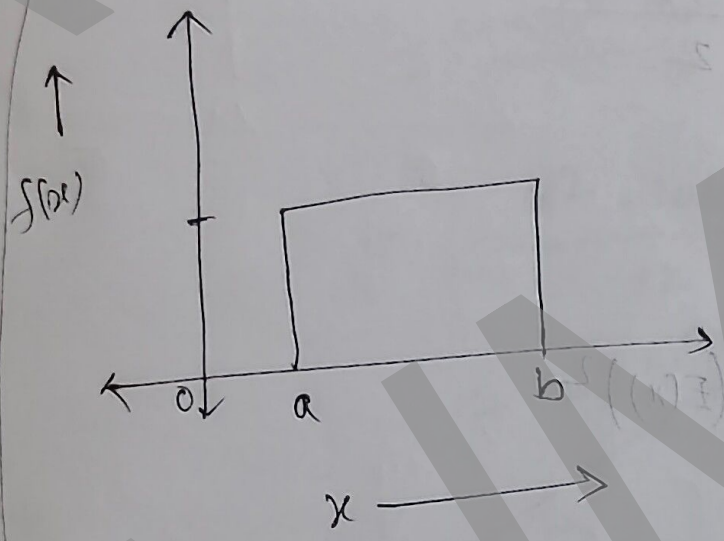
① Uniform Distribution:- / Rectangular Distribution:

PDF:-

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \le x \le b \\ 0, & \text{otherwise} \end{cases}$$

S.A.S / b

R.H.S / a



Notation:-

$X \sim U(a,b)$

$x \sim U(3,5)$

$f(x) = 1/2$

Mean:-

$$E(x) = \int_a^b x \cdot f(x)\, dx$$

$$= \int_a^b x \cdot \frac{1}{b-a}\, dx$$

$$= \frac{1}{b-a} \int_a^b x \, dx$$

$$= \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b$$

$$= \frac{1}{b-a} \left[\frac{b^2 - a^2}{2} \right]$$

$$= \frac{1}{(b-a)} \times \frac{(b+a)(b-a)}{2}$$

$$= \frac{a+b}{2}$$

$\therefore \text{var}(x) = E(x^2) - (E(x))^2$

$\therefore E(x^2) = \int_a^b x^2 \cdot f(x) \, dx$

$$= \int_a^b x^2 \cdot \frac{1}{b-a} \cdot dx$$

$$= \frac{1}{b-a} \left[\frac{x^3}{3} \right]_a^b$$

$$= \frac{1}{b-a} \left[\frac{b^3 - a^3}{3} \right]$$

$$= \frac{1}{b-a} \frac{(b-a)(a^2+ab+b^2)}{3}$$

$$= \frac{a^2+ab+b^2}{3}$$

$$\therefore \text{Var}(x) = \frac{a^2+ab+b^2}{3} - \frac{(a+b)^2}{4}$$

$$= \frac{a^2+ab+b^2}{3} - \frac{a^2+2ab+b^2}{4}$$

$$= \frac{4a^2+4ab+4b^2-3a^2-6ab-3b^2}{12}$$

$$= \frac{a^2-2ab+b^2}{12}$$

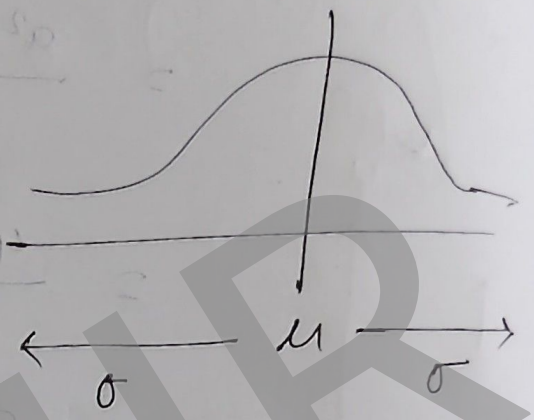
$$= \frac{(a-b)^2}{12}$$

Hence Proved

Normal distribution:-

PDF:-

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad -\infty < x < \infty$$



Notation:-

$$x \sim N(\mu, \sigma^2)$$

Mean:

$$E(x) = \mu$$

Variance:-

$$\text{var}(x) = \sigma^2$$

$$SD(x) = \sigma$$

L-A-Q 3

$$E(x) = \int_{-\infty}^{\infty} x \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

Let, $\frac{x-\mu}{\sigma} = z$

$\Rightarrow x - \mu = \sigma z$

$\Rightarrow x = \mu + \sigma z$

$\Rightarrow dx = \sigma\, dz$

$$E(x) = \int_{-\infty}^{\infty} (\mu + \sigma z) \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{z^2}{2}} \sigma dz$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} (\mu + \sigma z) \cdot e^{-\frac{z^2}{2}} \cdot dz$$

$$= \frac{1}{\sqrt{2\pi}} \left\{ \int_{-\infty}^{\infty} \mu \cdot e^{-\frac{z^2}{2}} dz + \int_{-\infty}^{\infty} \sigma z e^{-\frac{z^2}{2}} \cdot dz \right\}$$

$$= \frac{1}{\sqrt{2\pi}} \left\{ \int_{-\infty}^{\infty} \mu \cdot e^{-\frac{z^2}{2}} dz + 0 \right\}$$

$$\left[\because \sigma \int_{-\infty}^{\infty} z e^{-\frac{z^2}{2}} dz \right.$$

$$= \sigma \left\{ \int_{-\infty}^{0} z e^{-\frac{z^2}{2}} \cdot dz + \int_{0}^{\infty} z e^{-\frac{z^2}{2}} \cdot dz \right\}$$

$$= 0$$

$$\left. \text{Since } e \text{ this is an odd function} \right]$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \mu e^{-\frac{z^2}{2}} \cdot dz$$

$$= \frac{2\mu}{\sqrt{2\pi}} \int_{0}^{\infty} e^{-\frac{z^2}{2}} \cdot dz.$$

Let $\frac{z^2}{2} = y \Rightarrow z = \sqrt{2y}$

$\Rightarrow \frac{2z\,dz}{2} = dy$

$\Rightarrow dz = \frac{1}{z}\,dy$

$= \frac{1}{\sqrt{2y}}\,dy$

(b)

$\therefore E(x) = \frac{2\mu}{\sqrt{2\pi}} \int_0^\infty e^{-y} \frac{1}{\sqrt{2y}}\,dy$

$= \frac{\mu}{\sqrt{\pi}} \int_0^\infty e^{-y} y^{-1/2}\,dy$

$= \frac{\mu}{\sqrt{\pi}} \int_0^\infty e^{-y} y^{1/2 - 1}\,dy$

[Gamma function
$\int_0^\infty e^{-x} x^{n-1}\,dx = \lceil n$]

$= \frac{\mu}{\sqrt{\pi}} \lceil 1/2$

$= \frac{\mu}{\sqrt{\pi}} \times \sqrt{\pi} = \mu$ (proved)

L.A-9
5

$$E|X-\mu| = \int_{-\infty}^{\infty} |x-\mu| \cdot f(x)\,dx$$

$$\Rightarrow E|X-\mu| = \int_{-\infty}^{\infty} |x-\mu| \cdot \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \cdot dx$$

Let,

$$\frac{x-\mu}{\sigma} = z$$

$$\Rightarrow x-\mu = \sigma z$$

$$\Rightarrow x = \sigma z + \mu$$

$$\Rightarrow dx = \sigma\, dz.$$

Rough

$$|z| = \begin{cases} -z & -\infty \\ +z & 0 \end{cases}$$

$$\therefore E|x-\mu| = \int_{-\infty}^{\infty} |\sigma z| \cdot \frac{1}{\sigma\sqrt{2\pi}} \cdot e^{-\frac{z^2}{2}} \cdot \sigma\, dz$$

$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} |\sigma z| e^{-\frac{z^2}{2}} \cdot dz$$

$$= \frac{2}{\sqrt{2\pi}} \int_{0}^{\infty} \sigma z \cdot e^{-\frac{z^2}{2}} \cdot dz$$

$$= \frac{\sigma 2}{\sqrt{2\pi}} \int_0^{\infty} z \cdot e^{-\frac{z^2}{2}} \cdot dx$$

$\therefore$ Let $\frac{z^2}{2} = y$

$\Rightarrow z^2 = 2y$

$\Rightarrow 2z\,dz = 2\,dy$

$\Rightarrow z\,dz = dy$

$$\therefore E|x-\mu| = \frac{\sigma 2}{\sqrt{2\pi}} \int_0^{\infty} e^{-y}\,dy$$

$$= \frac{\sigma \times 2}{\sqrt{2\pi}} \left[-e^{-y} - (-e^0) \right]$$

$$= \frac{\sigma \times 2}{\sqrt{2\pi}} \left[0 - (-1) \right]$$

$$= \sigma \sqrt{\frac{2}{\pi}} \cdot 1$$

$$= \sigma \sqrt{\frac{2}{\pi}} \quad \text{(proved)}$$

Standard Normal Distribution:-

S.A-9
16

X: The heights of students $X \sim N(\mu = 170, \sigma^2 = (8)^2)$

$$P(X > 182) = P\left(\frac{X - \mu}{\sigma} > \frac{182 - 170}{8}\right)$$

$$= P(Z > 1.5)$$

$$= 1 - P(Z \leq 1.5)$$

$$= 1 - 0.9332$$

$$= 0.0668$$

L.A-9
19

Given,

$\mu = 1$ kg.

$\sigma^2 = (0.05)^2$

∴ X: The weights of packets of rice

$X \sim N(\mu = 1\text{kg}, \sigma^2 = (0.05)^2$

(i) X between 0.95 kg and 1.05 kg

$\therefore P(0.95 \leq X \leq 1.05)$

$= P(X \leq 1.05) - P(X \leq 0.95)$

$$= P\left(\frac{X-\mu}{\sigma} \leq \frac{1.05-1}{0.05}\right) - P\left(\frac{X-\mu}{\sigma} \leq \frac{0.95-1}{0.05}\right)$$

$$= P\left(\frac{x-\mu}{\sigma} \leq \frac{0.05}{0.05}\right) - P\left(\frac{x-\mu}{\sigma} \leq -\frac{0.05}{0.05}\right)$$

$$= P(Z \leq 1) - P(Z \leq -1)$$

$$= 0.8413 - \{1 - P(Z \leq 1)\}$$

$$= 0.8413 - \{1 - 0.8413\}$$

$$= 0.8413 - 0.1587$$

$$= 0.6826$$

(11) $P(x \leq 0.05) = 0.1587$

11/9.